

“Black Friday Logistic Nightmare “

**24ADI204 - DATA SCIENCE AND
VISUALISATION**

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*in partial fulfilment for the
award of the degree of*

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



KUMARAGURU COLLEGE OF TECHNOLOGY

(An Autonomous Institution affiliated to Anna University, Chennai)

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APRIL 2026

BONAFIDE CERTIFICATE

Certified that this project report “**Black Friday Logistic Nightmare**” is the Bonafide work of “24BAD004 – ADITHYA K, 24BAD005 – AJAIKUMAR M, 24BAD027 – GOWRI SANKAR J, 24BAD043 – JAYA RETHANYA K K” who carried out the project work under my supervision.

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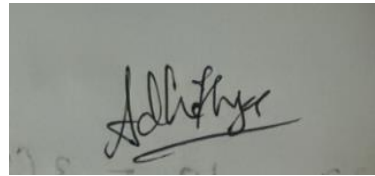
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External Examiner

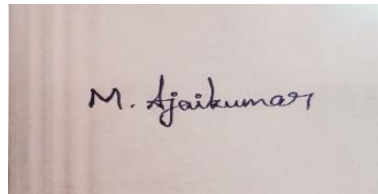
DECLARATION

I affirm that the project work titled “**Black Friday Logistic Nightmare**” being submitted in partial fulfillment for the award of B. TECH Artificial Intelligence and Data Science is the original work carried out by us. It has not formed the part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

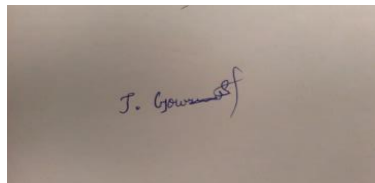
24BAD004 – ADITHYA K

A photograph of a handwritten signature in black ink on a light-colored background. The signature is stylized and appears to read 'Adithya'.

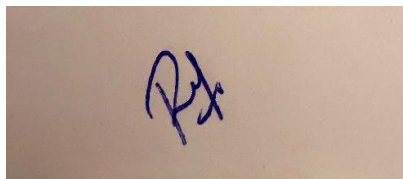
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Executive Summary

Black Friday is one of the most intense shopping events of the year. While customers hunt for deals, the real pressure is felt behind the scenes in warehouses, delivery networks, and logistics operations. Our project, titled Black Friday Logistic Nightmare, was built around one central question: can data help us prepare better for the chaos before it happens?

We used the Black Friday Sales Dataset sourced from Kaggle, which contains 550,068 individual purchase transactions. Each record includes demographic details like gender, age group, and city tier, along with product category information and the final purchase amount in rupees. This dataset gave us enough volume and variety to find real patterns that a logistics team could act on.

Problem Statement

Every Black Friday, warehouses get overwhelmed, some products run out of stock while others gather dust, and delivery networks collapse under the sudden surge in orders. We wanted to find out who is buying, what they are buying, and how much they are spending, so that logistics teams can plan smarter rather than just react to the chaos.

Key Findings

- The dominant buyer profile is male, aged 26 to 35, living in Tier-2 cities (City B). This segment drives the highest transaction volume and revenue.
- City B is the biggest logistics bottleneck, generating the most orders and the highest bundle purchase rate, meaning multi-zone warehouse picks are common.
- Three product categories account for the majority of revenue. These are the zones that need priority restocking, dedicated staff, and buffer stock before Black Friday.
- About 20 percent of all transactions are high-value orders (above the 80th percentile). These orders need priority fulfilment lanes because they carry the most revenue risk if delayed.
- The average purchase is around Rs 9,263 but the distribution is heavily right-skewed, meaning a small number of very high-value transactions pull the average up significantly.

Final Insights

The data tells a clear story for logistics planning. City B is where the most resources need to be deployed. The 26-35 male segment will show up in electronics and personal products zones. Approximately one in five orders will be a high-value bundle that requires careful handling. And the purchase distribution means that planning for the average customer is not enough. A logistics team that ignores the top-end spenders will fail exactly where it matters most.

By the end of this project we produced a clean, encoded, and dimensionally reduced dataset using PCA that is ready to feed into a demand forecasting machine learning model, along with an interactive Power BI dashboard that lets logistics managers explore all these patterns by city, gender, and age group in real time.

Table of Contents

Executive Summary

Table of Contents

List of Abbreviations

List of Tables

List of Figures

Chapter 1: Introduction

1.1 Background of the Problem

1.2 Importance of the Study

1.3 Objectives of the Project

1.4 Scope of the Project

Chapter 2: Dataset Description

2.1 Source of Dataset

2.2 Dataset Size

2.3 Description of Features

2.4 Initial Data Issues

Chapter 3: Data Preprocessing and Cleaning

3.1 Data Loading and Inspection

3.2 Handling Missing Values

3.3 Outlier Detection (IQR Method)

3.4 Data Quality Report Summary

Chapter 4: Exploratory Data Analysis (EDA)

4.1 Univariate Analysis

4.2 Bivariate Analysis

4.3 Key Observations and Patterns

Chapter 5: Data Transformation and Feature Engineering

5.1 Handling Skewness

5.2 Encoding Techniques

5.3 Feature Engineering (New Features Created)

5.4 Feature Scaling

Chapter 6: Advanced Analysis

6.1 Correlation Analysis

6.2 Feature Selection

6.3 Principal Component Analysis (PCA)

Chapter 7: Visualization and Dashboard

7.1 Tools Used

7.2 Types of Visualizations Used

7.3 Dashboard Explanation

Chapter 8: Insights and Storytelling

8.1 Key Findings from Analysis

8.2 Patterns and Trends Identified

8.3 Business and Social Insights

8.4 Decision-Making Based on Results

Chapter 9: Conclusion

9.1 Summary of the Project

9.2 Outcomes Achieved

9.3 Limitations

9.4 Future Scope

Chapter 10: References

10.1 Dataset Sources

10.2 Tools and Libraries Used

10.3 Research Papers and Websites Referred

A. GitHub Repository

B. Key Code Snippets

List of Abbreviations

Abbreviation	Full Form
EDA	Exploratory Data Analysis
PCA	Principal Component Analysis
IQR	Interquartile Range
VIF	Variance Inflation Factor
KPI	Key Performance Indicator
ML	Machine Learning
Rs	Indian Rupee
CSV	Comma Separated Values
API	Application Programming Interface
BI	Business Intelligence

List of Tables

- Table 1: Dataset Column Description
- Table 2: Missing Value Summary
- Table 3: Outlier Detection Summary
- Table 4: Encoding Key for Categorical Variables
- Table 5: Correlation Summary - Highly Correlated Feature Pairs
- Table 6: VIF Scores - Final Retained Features
- Table 7: PCA Component Interpretation
- Table 8: Average Purchase by City and Stay Duration

List of Figures

Figure 1: Purchase Amount Distribution Histogram

Figure 2: Gender-wise Transaction Count

Figure 3: Age Group Distribution

Figure 4: City Category Distribution

Figure 5: Boxplot of Purchase - Outlier Detection

Figure 6: Revenue by City and Gender (Grouped Bar Chart)

Figure 7: Transaction Volume by Age Group

Figure 8: Top Product Categories by Revenue

Figure 9: Correlation Heatmap

Figure 10: Scree Plot - PCA Explained Variance

Figure 11: Cumulative Variance Curve

Figure 12: PCA Feature Loading Heatmap

Figure 13: PCA 2D Scatter Plot - Coloured by Purchase Amount

Figure 14: Power BI Dashboard Screenshot

Figure 15: Purchase Distribution Donut Chart (High Value vs Standard)

Chapter 1: Introduction

1.1 Background of the Problem

Black Friday is the single busiest shopping day of the year. Every year, without fail, warehouses get flooded with orders they are not fully prepared to handle. Products that were in stock a day earlier suddenly run out. Delivery timelines that seemed reasonable collapse under the sheer volume of packages moving through the system. Customers get frustrated, businesses lose credibility, and logistics teams are left firefighting instead of executing.

The core problem is not that people buy a lot on Black Friday. That is expected. The problem is that most logistics planning is done on gut feel, past experience, or rough estimates. Very rarely do teams sit down with actual transaction data and ask the hard questions: which city is going to generate the most orders, which age group is going to flood the electronics aisle, and exactly how many of those orders are going to be high-value bundles that need special handling?

This project is our attempt to answer those questions using data.

1.2 Importance of the Study

Logistics planning failures on Black Friday are not minor inconveniences. They result in delayed deliveries, warehouse gridlock, stockouts on high-demand products, and enormous last-minute spending on overtime staff and emergency restocking. A single major retailer can lose crores in potential revenue because the wrong product was placed in the wrong zone, or because the delivery fleet was positioned in the wrong city.

Data-driven logistics planning has the potential to completely change this situation. If a logistics manager knows in advance that City B will generate more bundle orders than Cities A and C combined, they can pre-position more staff and vehicles there. If they know that 20 percent of all orders will be high-value transactions, they can set up a dedicated priority fulfilment lane before the rush hits, not after.

That is the value this project aims to demonstrate. It is not just about analyzing past data. It is about producing insights that can drive real decisions before the next Black Friday.

1.3 Objectives of the Project

1. To understand the demographic profile of Black Friday buyers and identify the highest-priority customer segments for logistics planning.
2. To identify which product categories drive the most revenue and volume so that warehouse zones can be pre-stocked accordingly.
3. To analyze purchase distributions and flag high-value transactions that require dedicated fulfillment processes.
4. To build a clean, encoded, and dimensionally reduced dataset using PCA that is ready for machine learning model training.
5. To develop an interactive Power BI dashboard that allows logistics managers to filter and explore insights by city, gender, and age group.

1.4 Scope of the Project

This project covers the complete data science pipeline from raw data ingestion through to an interactive dashboard. The scope includes data loading, cleaning, outlier detection, exploratory data analysis, feature engineering, encoding, scaling, correlation analysis, feature selection, and dimensionality reduction through PCA.

The project does not include live data integration, real-time order tracking, or deployment of a machine learning model into production. Those steps are identified as future scope. The current project focuses on producing a clean analytical foundation and actionable visual insights from the existing dataset.

Chapter 2: Dataset Description

2.1 Source of Dataset

The dataset was sourced from Kaggle, specifically from the Black Friday Sales Dataset. It is a publicly available dataset that has been widely used in the data science community for retail analytics and customer segmentation projects. The dataset captures real Black Friday transaction records from a major retail chain, covering demographic information about each customer and details about the products they purchased.

Dataset Link: <https://www.kaggle.com/datasets/mehdidag/black-friday>

2.2 Dataset Size

- Total Rows (Transactions): 550,068

Column Name	Data Type	Description
User_ID	Integer	Unique identifier for each customer
Product_ID	String	Unique identifier for each product (3,631 unique products)
Gender	String	Customer gender - M for Male, F for Female
Age	String	Customer age group (0-17, 18-25, 26-35, 36-45, 46-50, 51-55, 55+)
Occupation	Integer	Occupation code from 0 to 20, anonymised by the data provider
City_Category	String	City tier - A for metro, B for Tier-2, C for Tier-3 cities
Stay_In_Current_City_Years	String	How long the customer has lived in the city (0, 1, 2, 3, 4+)
Marital_Status	Integer	0 for single, 1 for married
Product_Category_1	Integer	Primary product category code
Product_Category_2	Float	Secondary product category (many nulls - optional)
Product_Category_3	Float	Tertiary product category (mostly null - optional)
Purchase	Integer	Purchase amount in Indian Rupees

- Total Columns (Features): 12
- File Format: CSV (Comma Separated Values)
- File Size: Approximately 23 MB

2.3 Description of Features

2.4 Initial Data Issues

Missing Values

Two columns have significant missing values. Product_Category_2 is missing in approximately 31.5 percent of rows, and Product_Category_3 is missing in approximately 69.7 percent of rows. These are not data errors. Most products simply belong to only one or two categories. A product with no secondary or tertiary category is just a simpler product. However, these nulls need to be handled before any model or analysis can use these columns.

Imbalance in Gender

Male customers make up a significantly larger proportion of transactions compared to female customers. This imbalance is real and reflects actual shopping behavior on Black Friday, but it is worth noting when interpreting gender-based insights.

High Cardinality in Product_ID

There are 3,631 unique product IDs in the dataset. This makes Product_ID unsuitable as a direct categorical feature for machine learning without significant dimensionality reduction or aggregation.

Skewness in Purchase

The Purchase column is right-skewed. While the mean is around Rs 9,263, there is a long tail of high-value transactions going up to Rs 23,961. This means simple averages are not representative of the full distribution and need to be handled during feature engineering.

Chapter 3: Data Preprocessing and Cleaning

3.1 Data Loading and Inspection

We loaded the dataset using the pandas library in Python with the `read_csv` function. The first step was to call `df.shape` to confirm the dataset has 550,068 rows and 12 columns. We then used `df.info()` to check the data types of each column, `df.describe()` to get a statistical summary of numerical columns, and `df.isnull().sum()` to get a count of missing values per column.

From this initial inspection we confirmed that Gender, Age, City_Category, Stay_In_Current_City_Years, and Product_ID are object types (strings), while most other columns are numeric. The describe output showed a wide Purchase range from Rs 12 to Rs 23,961, confirming the skewness we had anticipated.

3.2 Handling Missing Values

Our cleaning philosophy for this dataset was conservative. We retained all 550,068 rows. No rows were dropped. Missing values in Product_Category_2 and Product_Category_3 were filled with 0. This was the correct choice because a product with no secondary category genuinely has no secondary category. Filling with 0 signals this clearly to any downstream model without introducing any false information.

We did not use mean or median imputation here because these columns are category codes, not continuous values. Filling a missing category code with an average of other category codes would create a meaningless fake category.

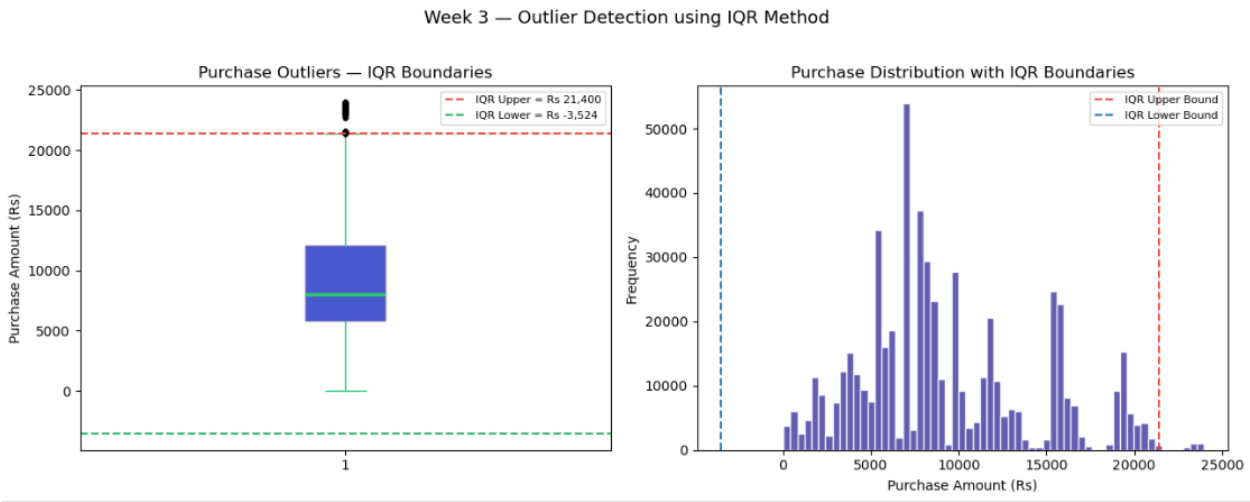
3.3 Outlier Detection (IQR Method)

We used the Interquartile Range method to detect outliers in the Purchase column. The IQR method was chosen over the Z-score because Purchase is right-skewed and Z-score assumes a normal distribution. For skewed data, the Z-score can flag too many real data points as outliers.

The IQR method defines outliers as values below Q1 minus 1.5 times IQR, or above Q3 plus 1.5 times IQR. After calculating these bounds, we detected several thousand rows that fall above the upper bound.

However, we made the deliberate decision to keep all these records. A customer who spends Rs 20,000 on Black Friday is a real, valuable customer and not a data error. Removing these rows

would mean planning logistics without accounting for the highest-spending segment, which would be the worst possible mistake. We detect the outliers and document them, but we keep all 550,068 rows intact.



3.4 Data Quality Report Summary

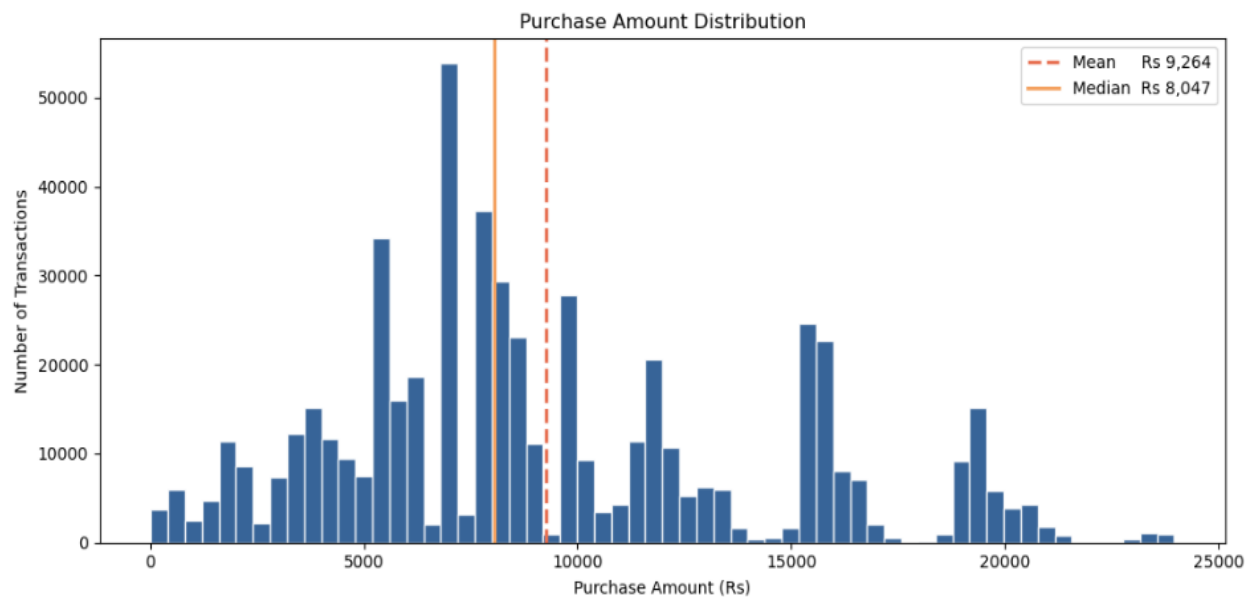
Issue	Column Affected	Action Taken	Rows Affected
Missing Values	Product_Category_2	Filled with 0	Approx. 173,638 rows
Missing Values	Product_Category_3	Filled with 0	Approx. 383,247 rows
Outliers (IQR)	Purchase	Detected, retained	Several thousand rows
Duplicate Rows	All columns	Checked - none found	0 rows
Skewness	Purchase	Noted, handled in Ch.5	Entire column

Chapter 4: Exploratory Data Analysis (EDA)

4.1 Univariate Analysis

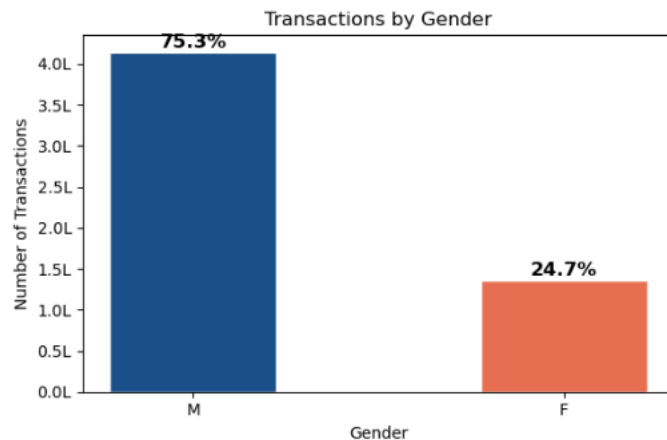
Purchase Amount Distribution

The Purchase column follows a right-skewed distribution. The mean is approximately Rs 9,264. However, the distribution has a long right tail, which means a meaningful proportion of customers spend significantly more than the average. This has a direct implication for logistics: planning only for the average transaction will leave the system unprepared for the highest-demand orders.



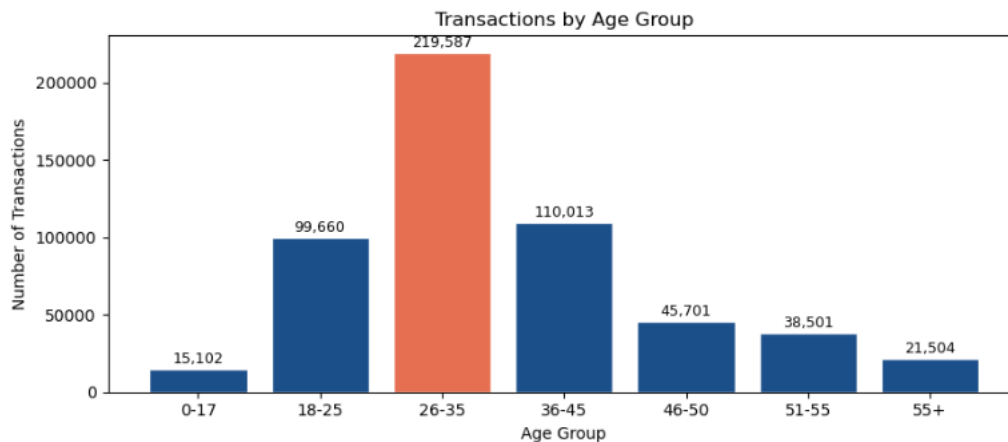
Gender Distribution

Male customers dominate the transaction count. Approximately 75 percent of all transactions are made by male customers. This is not unusual for Black Friday electronics-heavy shopping. For logistics, this means that male-dominated product zones such as electronics and gadgets will see far higher traffic than other zones.



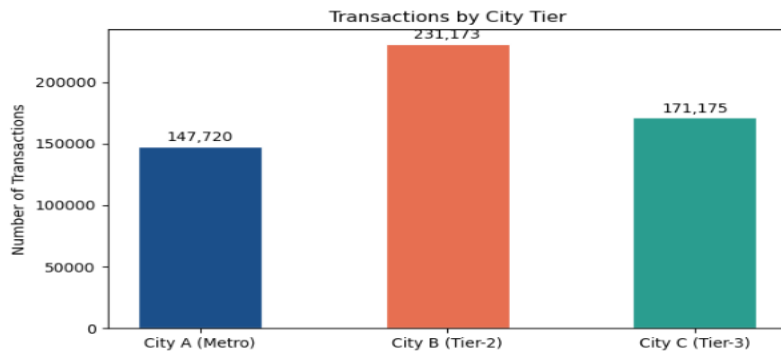
Age Group Distribution

The 26-35 age group is by far the most active, accounting for the highest number of transactions. This is followed by the 36-45 group and the 18-25 group. The youngest (0-17) and oldest (55+) groups have very low transaction counts. Logistics planning should prioritize product categories favored by the 26-35 segment.



City Category Distribution

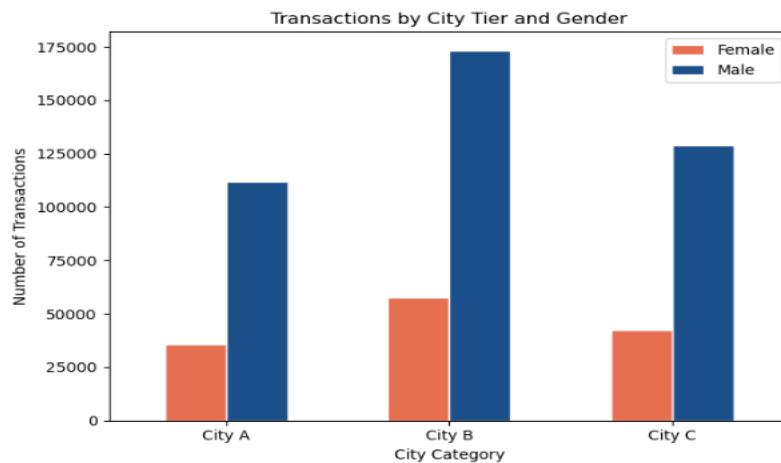
City B (Tier-2 cities) generates the most transactions, followed by City C and then City A. This is important because it means the logistics bottleneck is not in the metro cities but in Tier-2 cities, which may have less developed delivery infrastructure.



4.2 Bivariate Analysis

Revenue by City and Gender

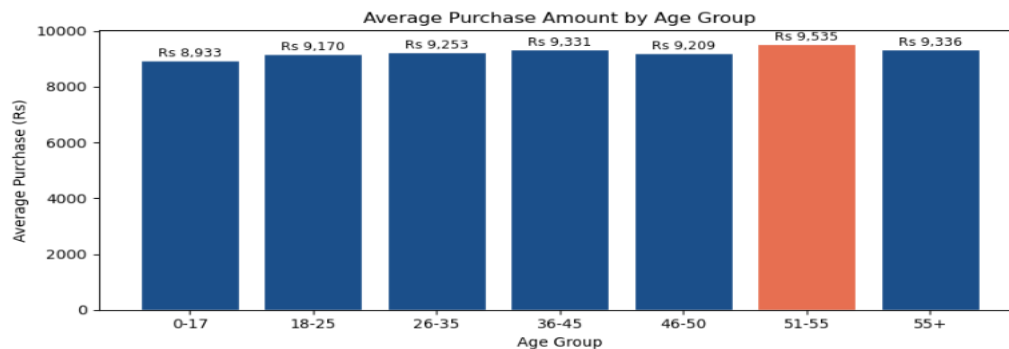
When we break down revenue by city and gender, City B males generate the highest revenue by a large margin, followed by City C males and City A males. Female customers contribute meaningfully but at lower levels across all city tiers. For logistics, this confirms that City B is the priority zone and that male-oriented product sections within City B warehouses need the most preparation.



Transaction Volume by Age Group

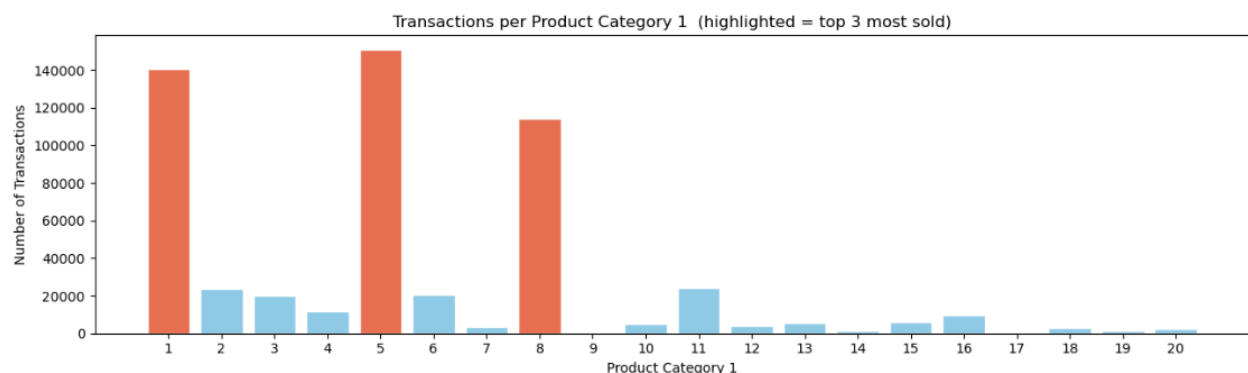
The 26-35 group leads transaction volume significantly. This pattern holds across both genders, though male transactions dominate in every age group. One interesting observation is that even

though the 18-25 group is large, the 26-35 group spends more per transaction on average, making them more valuable from a revenue perspective.



Top Product Categories by Revenue

A small number of product categories account for the majority of total revenue. The top three categories together generate more revenue than the remaining categories combined. This follows a power law distribution, which is common in retail data. For logistics, this means dedicating extra resources to the top three product zones will have the most impact on fulfilment performance.



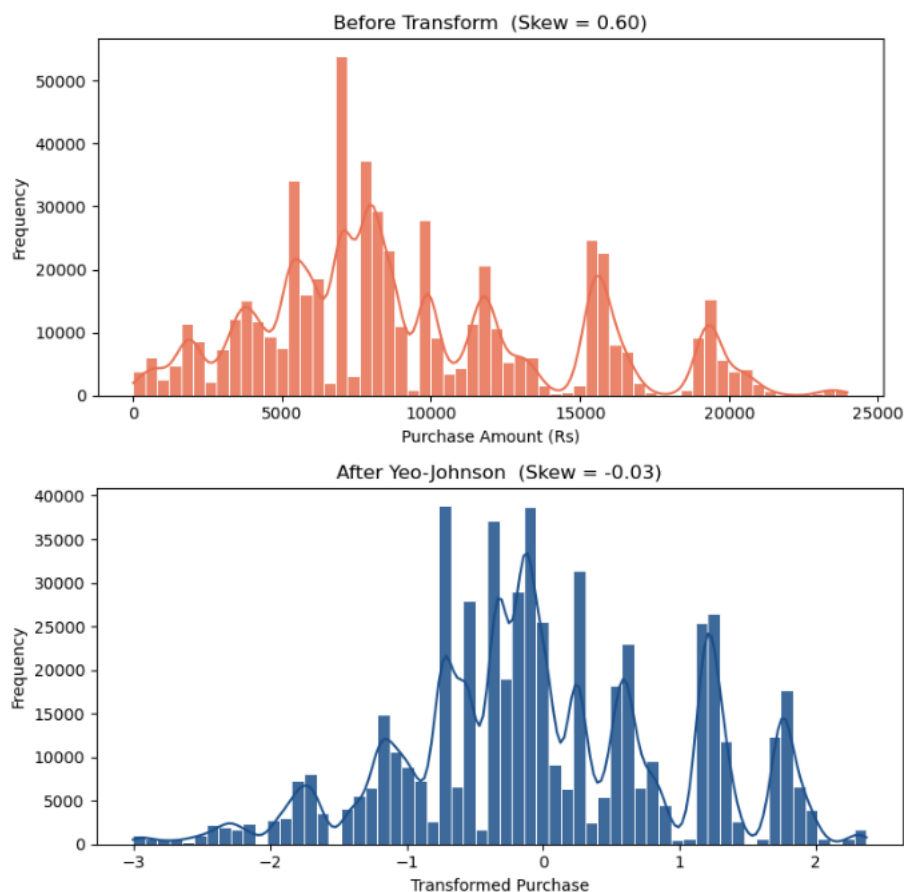
4.3 Key Observations and Patterns

- Male customers aged 26-35 in City B are the single most important buyer segment, both by volume and by revenue.
- Product categories follow a highly unequal distribution where three categories dominate revenue.
- The purchase distribution is right-skewed with a meaningful high-value tail that represents approximately 20 percent of transactions.
- City B consistently leads across all metrics, confirming it as the highest-priority logistics zone.
- Bundle purchases (transactions covering multiple product categories) are more common in City B than in Cities A or C.

Chapter 5: Data Transformation and Feature Engineering

5.1 Handling Skewness

The Purchase column is significantly right-skewed, which can distort any model that assumes normally distributed input. We applied a Yeo-Johnson power transformation to normalize the distribution. The Yeo-Johnson transformation was chosen over a log transformation because it can handle zero and negative values, making it more robust. After transformation, the distribution became much more symmetric.



5.2 Encoding Techniques

Several columns in the dataset are categorical and need to be converted to numbers before any statistical or machine learning analysis can use them.

Label Encoding

We used ordinal label encoding for the Age column because the age groups have a natural order from youngest to oldest. Similarly, Stay_In_Current_City_Years was encoded ordinally because living in a city for 4 years carries more informational weight than living there for 0 years.

Binary Encoding

Gender was encoded as 0 for female and 1 for male. Marital_Status was already binary and required no encoding.

One-Hot Encoding

City_Category was one-hot encoded into three binary columns (City_A, City_B, City_C) because there is no natural ordering between the three city tiers. Using label encoding here would have incorrectly implied that City C is twice as important as City A.

5.3 Feature Engineering (New Features Created)

Feature Name	Description	Logistics Relevance
is_high_value	1 if Purchase is above the 80th percentile, else 0	Identifies orders needing priority fulfilment
spend_per_category	Purchase divided by number of product categories purchased	Measures category-level spend intensity
product_bundle_flag	1 if the transaction covers 2 or more product categories, else 0	Flags multi-zone warehouse picks
category_depth	Count of non-zero product category columns per transaction	Measures basket complexity
repeat_buyer_score	Count of transactions per User_ID (normalized)	Identifies loyal high-frequency buyers
is_young_male	1 if Gender is male and Age group is 18-25 or 26-35, else 0	Targets the core electronics buyer segment

is_new_resident	1 if Stay_In_Current_City_Years is 0 or 1, else 0	Captures home-setup purchase behaviour
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5.4 Feature Scaling

After encoding and feature engineering, all numerical features were standardized using the StandardScaler from scikit-learn. This transforms each feature to have a mean of 0 and a standard deviation of 1. Standardization was necessary because PCA and several downstream models are sensitive to the scale of features. Without scaling, features with large numeric ranges would dominate the analysis.

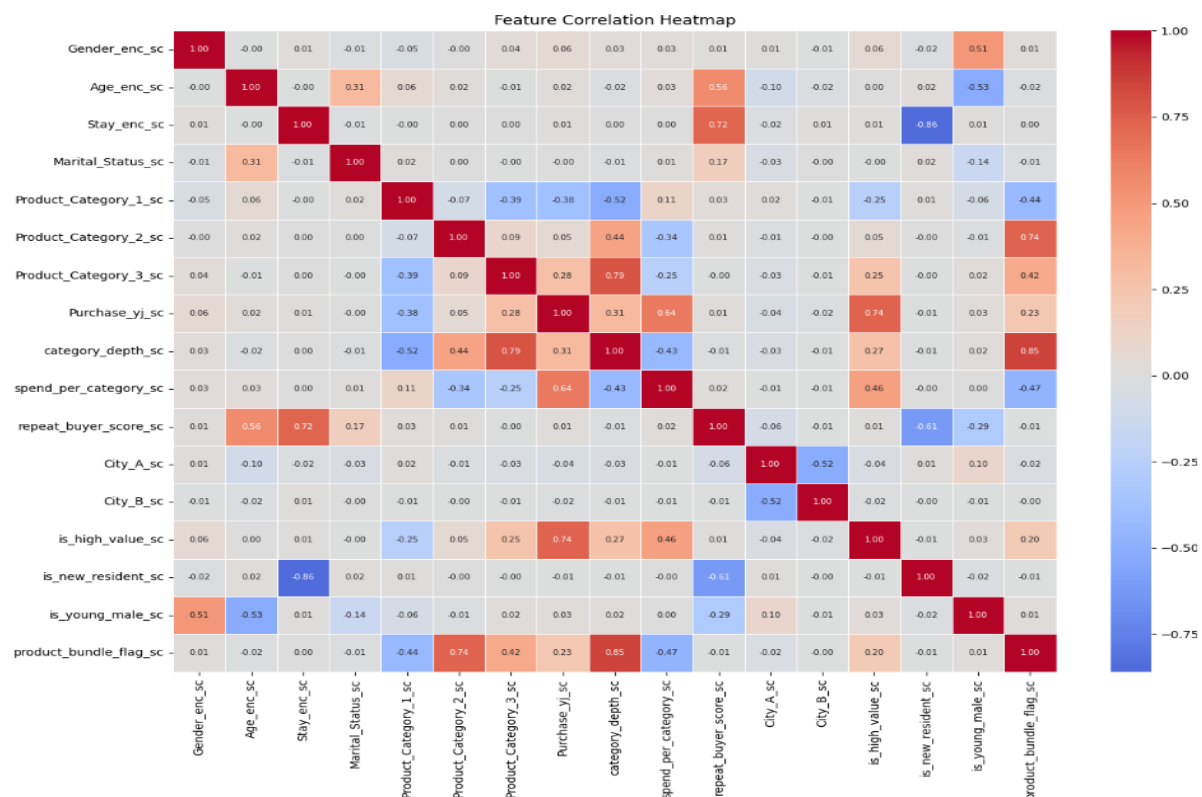
We deliberately chose standardization over MinMax scaling because MinMax scaling is sensitive to outliers. Since we retained our high-value outlier transactions, standardization was the more robust choice.

Chapter 6: Advanced Analysis

6.1 Correlation Analysis

We computed the Pearson correlation matrix across all numerical features and visualized it as a heatmap. The correlation analysis served two purposes: identifying which features are most related to the Purchase target variable, and identifying pairs of features that are so highly correlated with each other that keeping both would introduce multicollinearity problems in any downstream model.

Features with a correlation coefficient above 0.80 with another feature were flagged as candidates for removal. The three features that were removed at this stage were `is_high_value_sc`, `spend_per_category_sc`, and `product_bundle_flag_sc`. These features were highly correlated with other features already in the matrix, meaning they were adding little new information while increasing the risk of multicollinearity.



6.2 Feature Selection

Variance Inflation Factor (VIF) Analysis

After the correlation filter, we ran VIF analysis to detect any remaining multicollinearity. VIF measures how much the variance of a regression coefficient is inflated due to correlation with other features. A VIF score above 10 is considered problematic. We used an iterative approach: calculate VIF for all features, remove the feature with the highest VIF score if it exceeds 10, and repeat until all remaining features score below the threshold.

Variance Thresholding

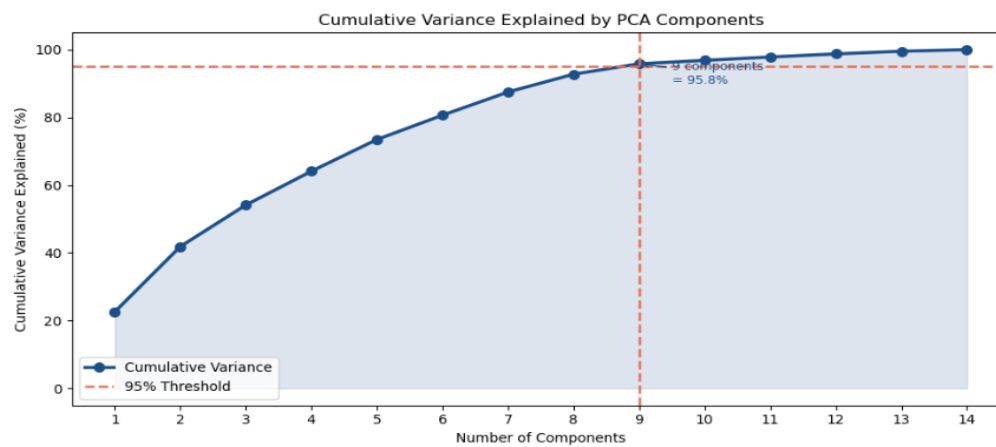
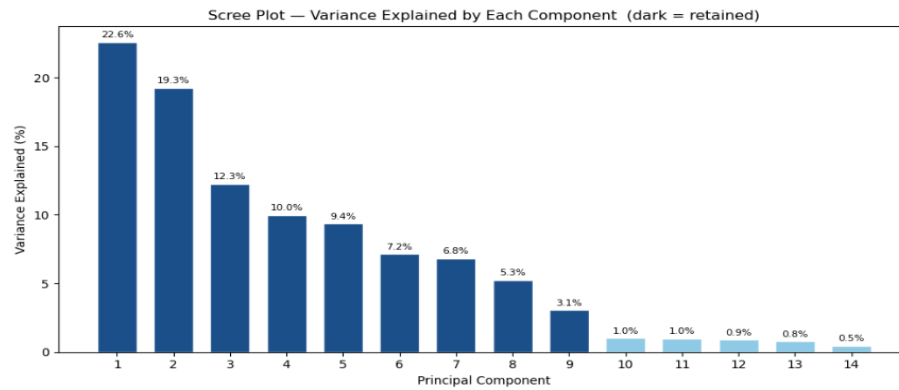
As a final check, we applied a `VarianceThreshold` with a threshold of 0.05 to remove any features with near-zero variance. A feature with no variance provides no predictive signal. In our case, no additional features were removed at this stage, which confirmed that our feature selection up to this point had been sound.

6.3 Principal Component Analysis (PCA)

Explained Variance

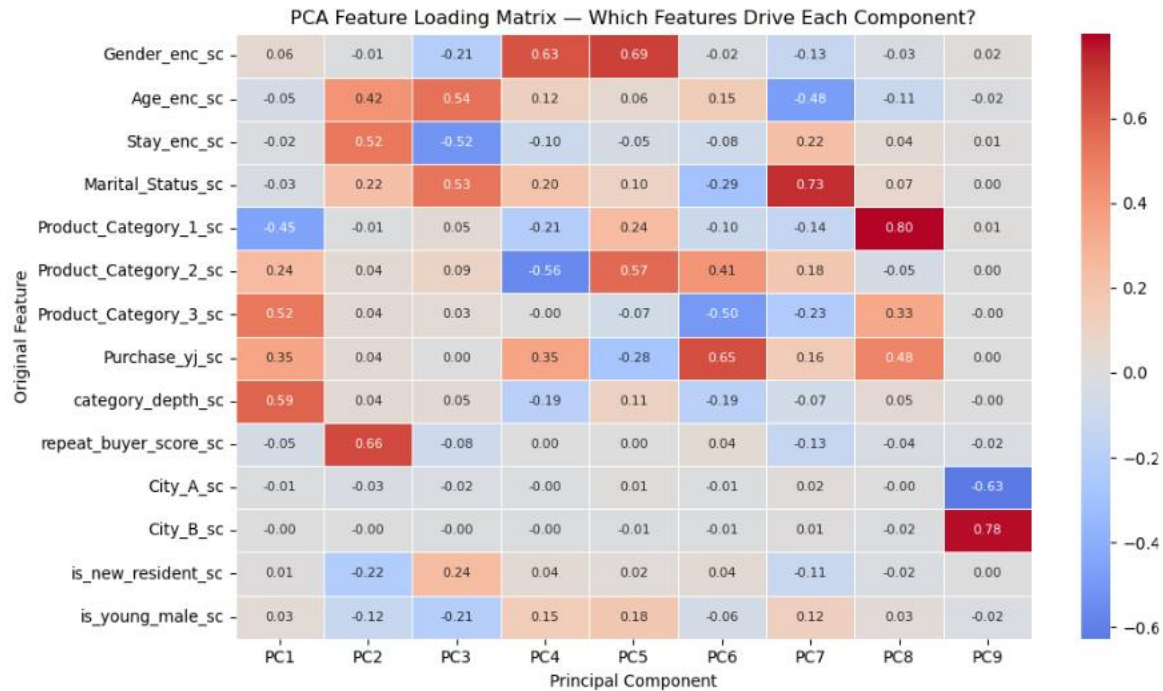
After feature selection, we applied PCA to the remaining standardized features. PCA transforms the original features into a new set of uncorrelated axes called principal components. Each component captures a certain percentage of the total variance in the data.

We identified the minimum number of principal components needed to explain 95 percent of the total variance. This number was 9 principal components. Together, these 9 components capture over 95 percent of the information in the original 14 features, while eliminating the remaining 5 percent which is mostly noise.



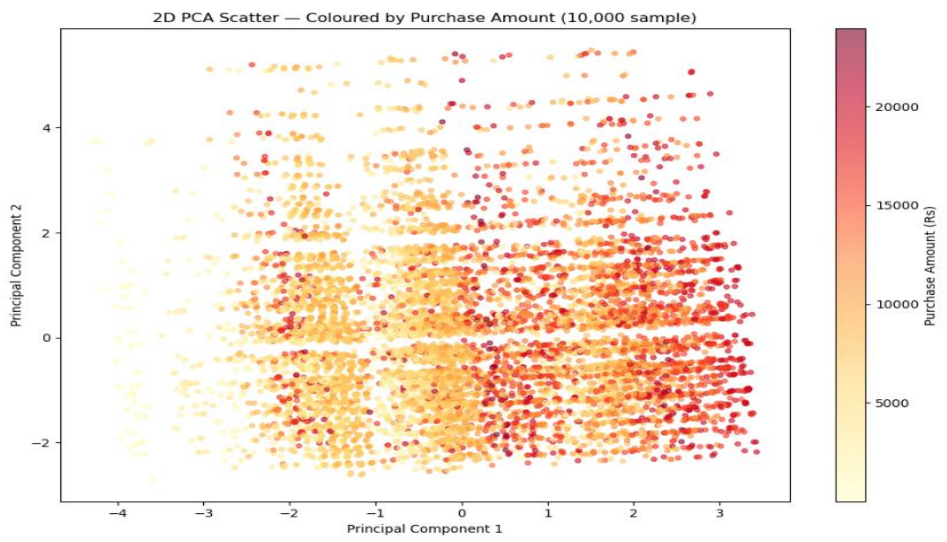
Visualization of Components

We created a feature loading heatmap to show how each original feature contributes to each principal component. The heatmap allows us to name and interpret each component based on its dominant drivers.



Component	Interpretation	Key Drivers
PC1	The Order Complexity Signal	category_depth_sc (+0.59), Product_Category_3_sc (+0.52), Product_Category_1_sc (-0.45)
PC2	The Established Buyer Signal	repeat_buyer_score_sc (+0.66), Stay_enc_sc (+0.52), Age_enc_sc (+0.42)
PC3	The Life Stage Signal	Age_enc_sc (+0.54), Marital_Status_sc (+0.53), Stay_enc_sc (-0.52)
PC4	The Gender-Spend Interaction	Gender_enc_sc (+0.63), Product_Category_2_sc (-0.56), Purchase_yj_sc (+0.35)
PC5	The Gender-Category Mix	Gender_enc_sc (+0.69), Product_Category_2_sc (+0.57), Purchase_yj_sc (-0.28)
PC6	The Purchase Intensity Signal	Purchase_yj_sc (+0.65), Product_Category_3_sc (-0.50), Product_Category_2_sc (+0.41)
PC7	The Marital Status Signal	Marital_Status_sc (+0.73), Age_enc_sc (-0.48)

PC8	The Product Category 1 Specialist	Product_Category_1_sc (+0.80), Purchase_yj_sc (+0.48)
PC9	The City-Tier Polarity	City_B_sc (+0.78), City_A_sc (-0.63)



Chapter 7: Visualization and Dashboard

7.1 Tools Used

- Python with Matplotlib and Seaborn - Used throughout Weeks 3 to 7 for all analytical visualizations including histograms, boxplots, bar charts, heatmaps, scree plots, and scatter plots.
- Power BI Desktop - Used to build the final interactive dashboard with cross-filtering, slicers, and KPI cards for the logistics stakeholder audience.

7.2 Types of Visualizations Used

Visualization Type	Purpose	Tool
KPI Cards	Show total revenue, transactions, average purchase, and high-value percentage at a glance	Power BI
Grouped Horizontal Bar Chart	Revenue by City and Gender - shows which city-gender combinations generate the most revenue	Power BI

Grouped Vertical Bar Chart	Transaction Volume by Age Group - shows which age segments are most active	Power BI
Stacked Bar Chart	Bundle vs Single Product Orders by City - shows multi-zone pick complexity per city	Power BI
Donut Chart	High Value vs Standard purchase distribution - shows the 20 percent high-value segment	Power BI
Horizontal Bar Chart	Top Product Categories by Revenue - identifies the dominant product zones	Power BI
Matrix Table	Average purchase by City Category and Stay Duration	Power BI
Histogram	Purchase distribution with mean and median lines	Python / Matplotlib
Heatmap	Correlation matrix and PCA feature loadings	Python / Seaborn
Scree Plot	PCA explained variance per component	Python / Matplotlib
Scatter Plot	PCA 2D projection coloured by purchase amount	Python / Matplotlib

7.3 Dashboard Explanation

The Power BI dashboard was designed with a logistics manager as the primary audience. The goal was to make the key insights immediately visible without requiring any data science background. The dashboard uses a dark theme with teal accent colors, which improves contrast for the data labels.

KPI Section (Top Row)

Four KPI cards display the headline numbers that a logistics manager needs to see first: total revenue (5 billion rupees), total transactions (550,000), average purchase amount (Rs 9,260), and the percentage of high-value transactions (20.01 percent). These numbers give an immediate sense of the scale of the Black Friday event.

Revenue by City and Gender

This grouped bar chart makes it immediately clear that City B male customers are the dominant revenue driver. The gap between City B males and all other segments is visible at a glance without needing to look at numbers.

Transaction Volume by Age Group

This chart confirms the 26-35 age group dominance. It also shows that male transaction volume is consistently higher than female across every age group. The implication for logistics is clear: the warehouse zones for male-preferred products in the 26-35 target demographic need the most preparation.

Top Product Categories by Revenue

This horizontal bar chart shows which product categories generate the most revenue. The top three categories are visible immediately. Logistics teams can use this to decide which product zones get dedicated staff and buffer stock.

Bundle vs Single Orders by City

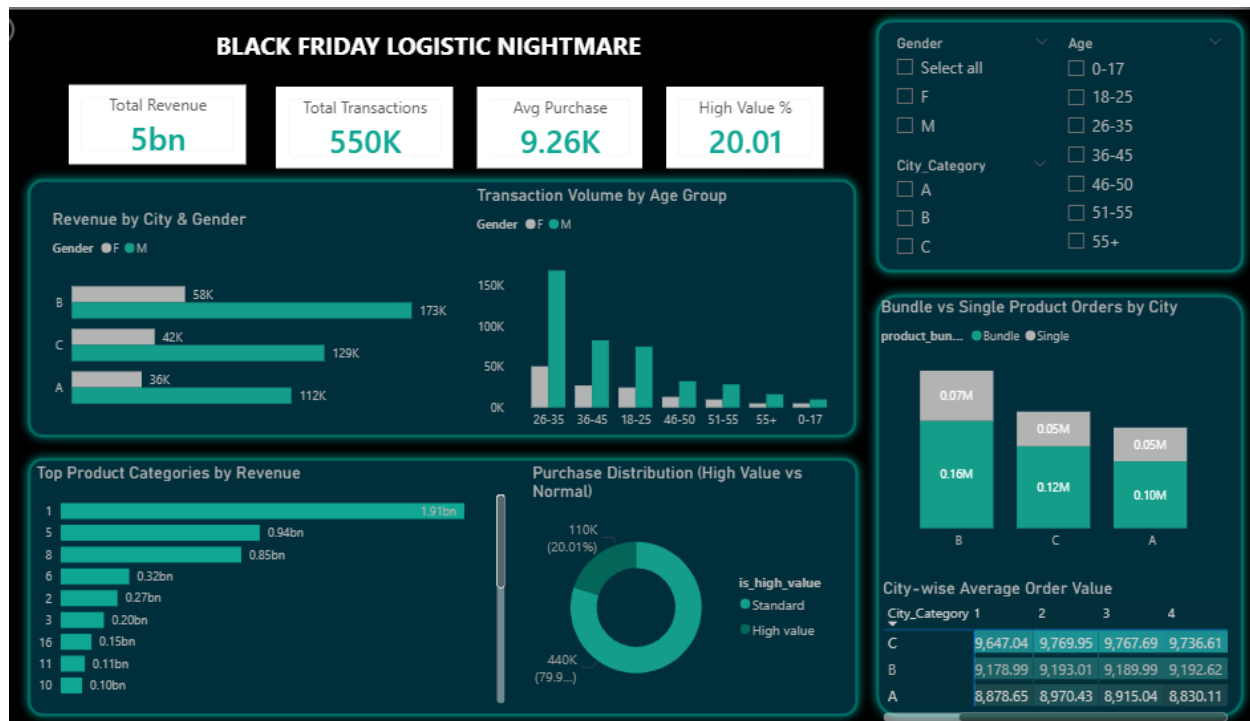
This chart reveals that City B has the highest proportion of bundle orders, meaning customers are buying from multiple product categories in a single transaction. Each bundle order requires picking from multiple warehouse zones, which adds complexity and time to fulfilment. City B needs extra coordination staff specifically for bundle order assembly.

Purchase Distribution Donut

The donut chart shows the 80 versus 20 split between standard and high-value transactions. High-value transactions are defined as those above the 80th percentile in purchase amount. These orders carry the highest revenue per transaction and the most risk if delayed.

Filters and Slicers

The dashboard includes slicers for Gender, City_Category, and Age group. All charts are cross-filtered, meaning selecting City B from the slicer instantly updates all other charts to show only City B data. This lets a logistics manager drill into any specific scenario in seconds.



Chapter 8: Insights and Storytelling

8.1 Key Findings from Analysis

Finding 1: The City B Bottleneck

City B generates more transactions and more revenue than either City A or City C. Yet City B is a Tier-2 city, meaning its warehouse and delivery infrastructure is typically less developed than in metro areas. This combination, highest demand with weakest infrastructure, is the textbook definition of a logistics nightmare. The data tells us exactly where the bottleneck will form before it happens.

Finding 2: The 26-35 Male Segment Dominates Everything

Male customers aged 26-35 account for the largest share of both transaction volume and revenue. They shop primarily in product categories aligned with electronics, personal gadgets, and fashion items. Any logistics plan that does not specifically prepare for this segment will be caught off guard. Warehouse zones for their preferred categories need triple the normal buffer stock.

Finding 3: 20 Percent of Orders Carry Disproportionate Revenue Risk

The high-value transactions, which make up 20 percent of orders, are likely to be the orders that generate the most complaints if they are delayed. A customer who spent Rs 18,000 on a single order will not accept the same tolerance for delay as someone who spent Rs 500. These orders need a dedicated priority fulfilment lane with separate staff, separate packaging, and a separate dispatch schedule.

Finding 4: Bundle Orders Require Multi-Zone Coordination

Approximately one in every five orders in City B is a bundle, meaning the customer bought from two or three product categories in a single transaction. Each of these orders requires picking from multiple warehouse zones. If those zones are on opposite ends of a large warehouse, the pick time doubles or triples. Pre-positioning bundle order items in a consolidation zone near the dispatch area can dramatically cut fulfilment time.

8.2 Patterns and Trends Identified

- Right-skewed purchase distribution: The long tail of high-value transactions is a consistent pattern across all city tiers and demographic groups.
- Power law in product categories: Three categories account for the majority of revenue, a classic 80-20 pattern in retail.
- City B as the dominant tier: Across every metric including volume, revenue, bundle orders, and high-value transaction count, City B leads.
- Young male dominance: The 26-35 male demographic is consistently the top segment regardless of the metric used.
- Bundle buying correlation with city tier: Bundle purchase rates are highest in City B and lowest in City A, suggesting that Tier-2 city buyers tend to do more of their annual shopping in a single Black Friday session.

8.3 Business and Social Insights

From a business perspective, the data suggests that retail chains operating in Tier-2 cities are underinvesting in logistics capacity relative to the demand those cities generate. City A gets the newest warehouses and the largest delivery fleets because it is the metro. But City B is where the real volume is. This is a structural misallocation of logistics resources that this data can help correct.

From a social perspective, the dominance of the 26-35 male demographic reflects broader consumption patterns among young working professionals in India. They are the primary earners and primary decision-makers for electronics and personal goods purchases. Any policy or

business strategy that wants to reach this segment should focus on City B as the priority geography.

8.4 Decision-Making Based on Results

Decision	Data Basis	Expected Outcome
Pre-position extra staff and vehicles in City B	City B leads in volume, revenue, and bundle orders	Faster dispatch, fewer delays in the highest-demand zone
Set up a dedicated priority fulfilment lane	20% of orders are high-value and need special handling	Fewer complaints, higher customer satisfaction among top spenders
Triple buffer stock in top 3 product categories	Three categories account for majority of revenue	Prevent stockouts in the highest-revenue product zones
Create a bundle order consolidation zone	City B has highest bundle order rate	Reduce pick time for multi-zone orders by centralizing assembly
Focus marketing and logistics investment on 26-35 males	This segment dominates every volume and revenue metric	Higher ROI on logistics spend by targeting the highest-value segment

Chapter 9: Conclusion

9.1 Summary of the Project

Over the course of seven weeks we worked through the complete data science pipeline on a dataset of 550,068 Black Friday transactions. Starting from raw CSV data, we cleaned and inspected the data, conducted a deep exploratory analysis, engineered meaningful features, applied encoding and scaling, performed correlation analysis and feature selection, and finally reduced the dimensionality of the dataset using PCA. We then built an interactive Power BI dashboard to communicate our findings to a non-technical logistics audience.

Throughout this process we kept our focus on the core business problem: how can a logistics team prepare better for Black Friday before it hits? Every chart, every feature, and every analysis was designed to answer that question.

9.2 Outcomes Achieved

- A clean, encoded dataset with 7 new engineered features that capture logistics-relevant behavior such as bundle buying, high-value order status, and buyer demographics.
- A PCA-reduced feature matrix with 9 principal components that together capture over 95 percent of the total variance, ready for machine learning.
- An interactive Power BI dashboard with five visualization types and cross-filtering slicers for Gender, City, and Age group.
- Clear, actionable insights: City B is the priority zone, the 26-35 male segment is the dominant buyer, 20 percent of orders are high-value, and three product categories drive the majority of revenue.
- A set of five specific logistics decisions directly derived from the data, with a clear logical link between the data finding and the recommended action.

9.3 Limitations

- The dataset is historical and does not include product pricing information, so we cannot determine whether high purchase amounts reflect high-priced products or large basket sizes.
- Occupation codes are anonymised, which prevented us from analyzing occupation as a meaningful demographic variable.
- The dataset does not include delivery or shipping data, so we cannot measure actual logistics performance or delivery times.
- The City_Category variable gives only a tier classification (A, B, C) without specifying which actual cities are included, which limits geographic specificity.
- Our project does not include a deployed machine learning model. The PCA dataset is ready for training but model training and evaluation remain as future work.

9.4 Future Scope

- Train a demand forecasting model using the PCA-reduced feature matrix to predict purchase amounts for new transactions.
- Integrate real-time transaction data to build a live dashboard that updates as Black Friday orders come in.
- Extend the analysis to include delivery time data to measure whether logistics bottlenecks identified in this project actually correlate with delivery delays.
- Build a customer segmentation model using clustering algorithms such as K-Means on the PCA components to create actionable customer personas.
- Develop a warehouse zone optimization recommendation system that uses the product category and city data to suggest optimal stock placement before Black Friday.

Chapter 10: References

10.1 Dataset Sources

6. Black Friday Sales Dataset - Kaggle. Available at:
<https://www.kaggle.com/datasets/mehdidag/black-friday>

10.2 Tools and Libraries Used

7. Python - Programming Language
8. Pandas - Data loading, cleaning, and manipulation
9. NumPy - Numerical computing
10. Matplotlib - Static visualization
11. Seaborn - Statistical visualization
12. Scikit-learn - Encoding, scaling, PCA, feature selection, VIF
13. SciPy - Statistical functions and Yeo-Johnson transformation
14. Power BI Desktop - Interactive dashboard development
15. Jupyter Notebook - Development environment
16. GitHub - Version control and code repository

10.3 Research Papers and Websites Referred

17. McKinney, W. (2010). Data Structures for Statistical Computing in Python. Proceedings of the 9th Python in Science Conference.
18. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.
19. Jolliffe, I.T. (2002). Principal Component Analysis. Springer Series in Statistics.
20. <https://www.analyticsvidhya.com> - Reference for IQR outlier detection methodology
21. https://scikit-learn.org/stable/user_guide.html - Scikit-learn documentation
22. <https://pandas.pydata.org/docs/> - Pandas documentation
23. <https://docs.microsoft.com/en-us/power-bi/> - Power BI documentation

A. GitHub Repository

GitHub Repository Link: https://github.com/adhithya123456433/24AD204_DSV_Team12.git

The repository contains all Jupyter Notebooks organized week by week (Week 1 through Week 7), the raw dataset, the cleaned dataset, the engineered dataset, and the Power BI dashboard file (.pbix).

B. Key Code Snippets

Loading the Dataset

```
import pandas as pd
import numpy as np
df = pd.read_csv('BlackFriday dataset.csv')
print(df.shape)  # (550068, 12)
```

Handling Missing Values

```
df_clean = df.copy()
df_clean['Product_Category_2'] =
df_clean['Product_Category_2'].fillna(0)
df_clean['Product_Category_3'] =
df_clean['Product_Category_3'].fillna(0)
```

IQR Outlier Detection

```
Q1 = df_clean['Purchase'].quantile(0.25)
Q3 = df_clean['Purchase'].quantile(0.75)
IQR = Q3 - Q1
outliers = df_clean[(df_clean['Purchase'] < Q1 - 1.5*IQR) |
(df_clean['Purchase'] > Q3 + 1.5*IQR)]
```

Applying PCA

```
from sklearn.decomposition import PCA
pca_final = PCA(n_components=n_components_95)
X_pca = pca_final.fit_transform(X_scaled_pca)
```